HW44 – Spring MVC Short Answer

1. List all annotations learned

Common annotations include: @Controller, @RequestMapping, @GetMapping, @PostMapping, @Autowired, @Service, @Repository, @Component, @Entity, @Table, @Id, @GeneratedValue, @Configuration, @Bean. These annotations help Spring manage objects and request routing.

2. Tight coupling vs Loose coupling & Spring IOC

Tight coupling means classes depend directly on concrete implementations, making changes difficult. Loose coupling means classes depend on interfaces. Spring IOC (Inversion of Control) creates and manages objects automatically, injecting dependencies to reduce coupling.

3. What is MVC pattern?

MVC stands for Model-View-Controller. Model handles data and business logic, View displays UI (JSP), Controller handles requests and coordinates Model and View.

4. What is Front Controller?

Front Controller is a design pattern where all requests go through one central servlet. In Spring MVC, DispatcherServlet acts as the Front Controller.

5. DispatcherServlet explanation

DispatcherServlet receives all HTTP requests, finds the correct controller using HandlerMapping, executes controller methods, and returns the view through ViewResolver.

6. What is JSP, Model and View?

JSP is a Java-based template technology used to generate dynamic HTML. Model contains data sent from controller. View is the page shown to users.

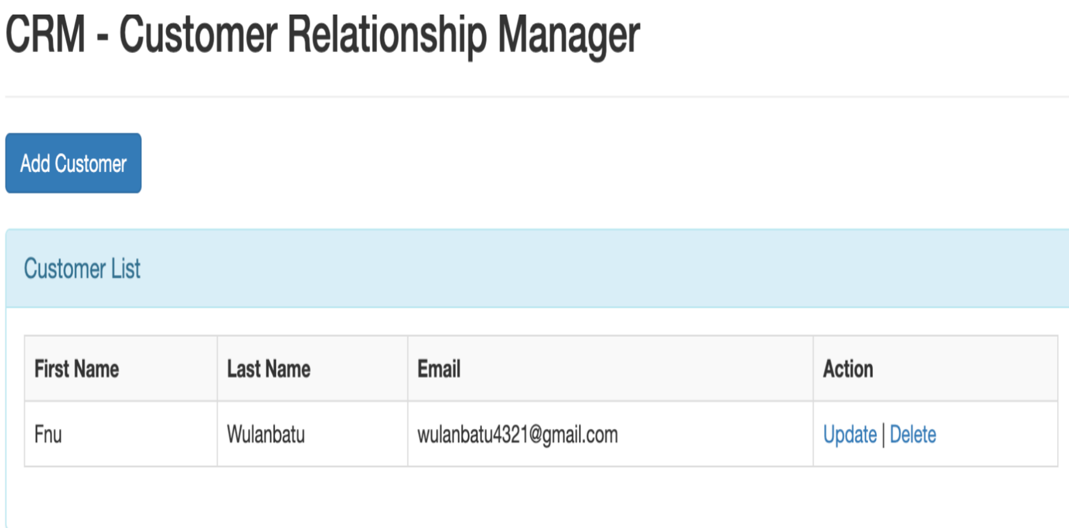
7. Servlet and Servlet Container

Servlet is a Java class that handles HTTP requests. Servlet container manages servlets lifecycle. Other servlet containers include Jetty, Undertow, WebLogic, WebSphere.

8. Hands-on summary

The project was cloned and deployed locally. Tomcat 9 was configured manually. MySQL was set up using Docker. APIs were tested successfully using browser and Postman.

Application Running Screenshot:



CRM - Customer Relationship Manager

Add Customer

Customer List

First Name	Last Name	Email	Action
Fnu	Wulanbatu	wulanbatu4321@gmail.com	Update | Delete